Aim: To create a data report analysis using data-profiling

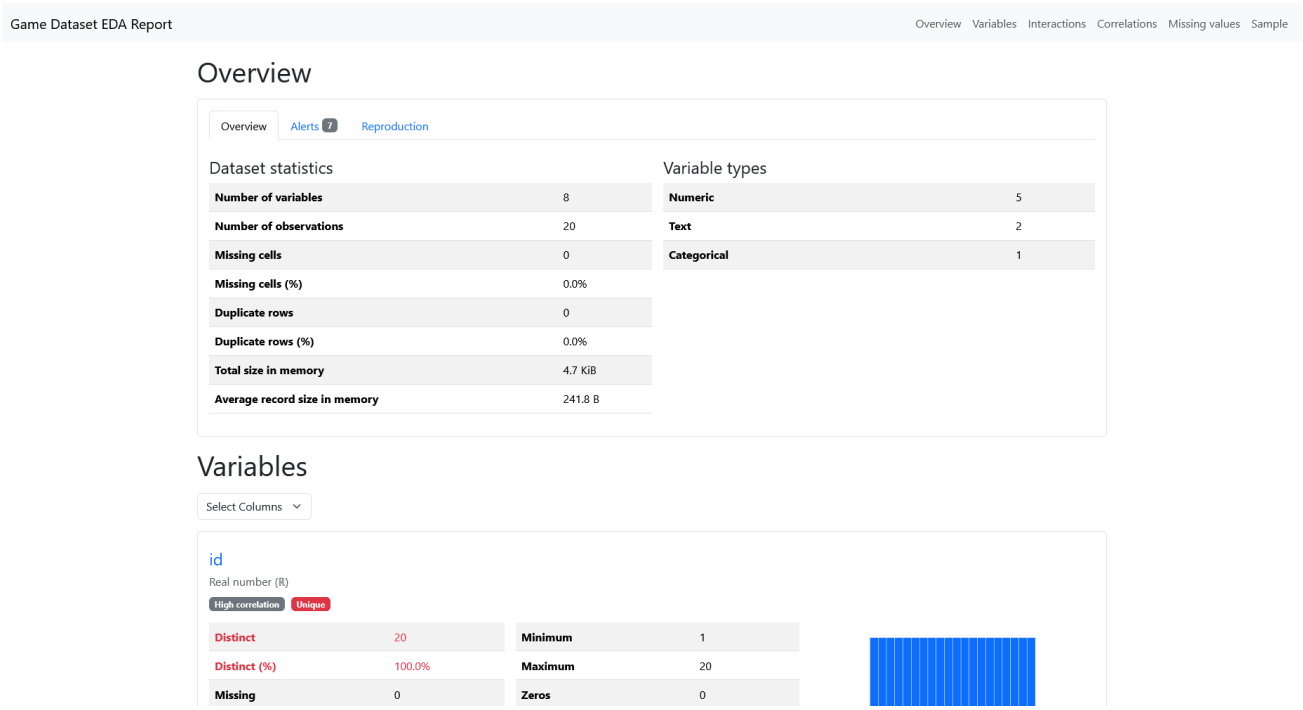
Code:

```
import pandas as pd
from ydata_profiling import ProfileReport

# Load dataset
df = pd.read_csv("Untitled.csv")

# Generate EDA report
ProfileReport(df, title="Game Dataset EDA Report",
explorative=True).to_file("game_eda_report.html")
```

Output:



Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Overview

Overview | Alerts 7 | Reproduction

Alerts

genre is highly overall correlated with peak_players	High correlation
id is highly overall correlated with peak_players	High correlation
peak_players is highly overall correlated with genre and 1 other fields	High correlation
id has unique values	Unique
name has unique values	Unique
sales_millions has 1 (5.0%) zeros	Zeros
peak_players has 10 (50.0%) zeros	Zeros

Variables

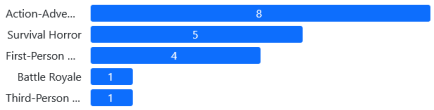
genre

genre

Categorical

High correlation

Distinct	6
Distinct (%)	30.0%
Missing	0
Missing (%)	0.0%



Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Variables

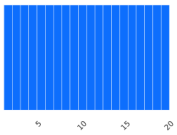
Select Columns

id

Real number (ℝ)

High correlation Unique

Distinct	20	Minimum	1
Distinct (%)	100.0%	Maximum	20
Missing	0	Zeros	0
Missing (%)	0.0%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	10.5	Memory size	292.0 B



More details

name

Text

Unique

Distinct	20
Distinct (%)	100.0%
Missing	0
Missing (%)	0.0%
Memory size	1.5 KiB

Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

developer

Text

Distinct	10
Distinct (%)	50.0%
Missing	0
Missing (%)	0.0%
Memory size	1.3 KiB

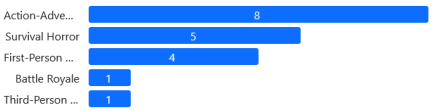
More details

genre

Categorical

High correlation

Distinct	6
Distinct (%)	30.0%
Missing	0
Missing (%)	0.0%
Memory size	1.4 KiB



More details

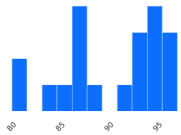
Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

metacritic_score

Real number (ℝ)

Distinct	11	**Minimum**	80
Distinct (%)	55.0%	**Maximum**	97
Missing	0	**Zeros**	0
Missing (%)	0.0%	**Zeros (%)**	0.0%
Infinite	0	**Negative**	0
Infinite (%)	0.0%	**Negative (%)**	0.0%
Mean	90.2	**Memory size**	292.0 B



More details

Statistics Histogram Common values Extreme values

Quantile statistics

Minimum	80
5-th percentile	80
Q1	87
median	92
Q3	94
95-th percentile	97
Maximum	97
Range	17
Interquartile range (IQR)	7

Descriptive statistics

Standard deviation	5.2475558
Coefficient of variation (CV)	0.058176894
Kurtosis	-0.62325372
Mean	90.2
Median Absolute Deviation (MAD)	4.5
Skewness	-0.5674211
Sum	1804
Variance	27.536842
Monotonicity	Not monotonic

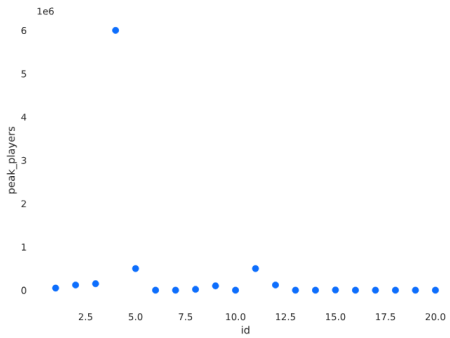
Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Interactions

id release_year metacritic_score sales_millions peak_players

peak_players id release_year metacritic_score sales_millions



Correlations

Auto

Heatmap Table

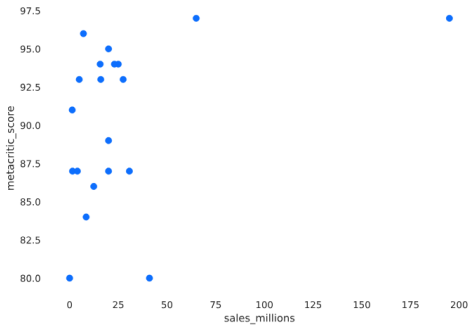
Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Interactions

id release_year metacritic_score sales_millions peak_players

peak_players id release_year metacritic_score sales_millions



Correlations

Auto

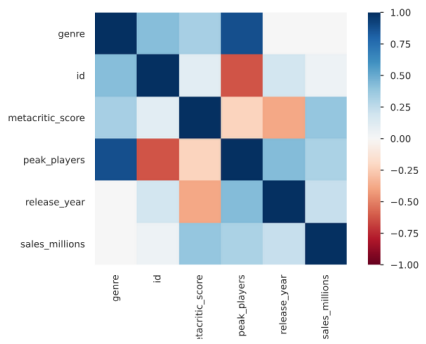
Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Correlations

Auto

Heatmap Table



Missing values

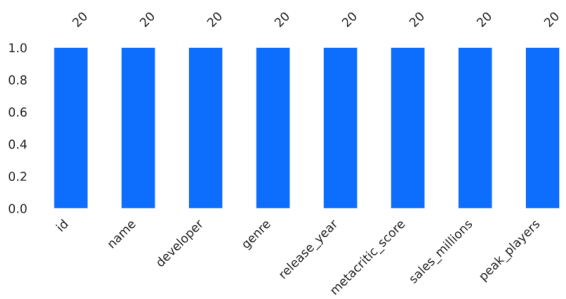
Count Matrix

Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample

Missing values

Count Matrix



A simple visualization of nullity by column.

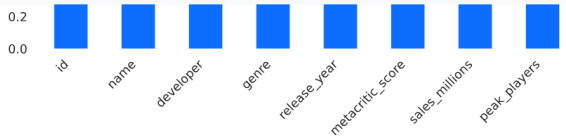
Sample

First rows Last rows

	id	name	developer	genre	release_year	metacritic_score	sales_millions	peak_players
0	1	Call of Duty 4: Modern Warfare	Infinity Ward	First-Person Shooter	2007	94	15.70	50000
1	2	Call of Duty: Modern Warfare 2	Infinity Ward	First-Person Shooter	2009	94	25.00	120000

Game Dataset EDA Report

Overview Variables Interactions Correlations Missing values Sample



A simple visualization of nullity by column.

Sample

First rows Last rows

	id	name	developer	genre	release_year	metacritic_score	sales_millions	peak_players
0	1	Call of Duty 4: Modern Warfare	Infinity Ward	First-Person Shooter	2007	94	15.70	50000
1	2	Call of Duty: Modern Warfare 2	Infinity Ward	First-Person Shooter	2009	94	25.00	120000
2	3	Call of Duty: Black Ops	Treyarch	First-Person Shooter	2010	87	30.70	150000
3	4	Call of Duty: Warzone	Infinity Ward	Battle Royale	2020	80	0.00	6000000
4	5	Call of Duty: Modern Warfare (2019)	Infinity Ward	First-Person Shooter	2019	80	41.00	500000
5	6	Resident Evil 2	Capcom	Survival Horror	1998	93	4.96	0
6	7	Resident Evil 4	Capcom	Survival Horror	2005	96	7.10	0
7	8	Resident Evil 7: Biohazard	Capcom	Survival Horror	2017	86	12.40	20000
8	9	Resident Evil Village	Capcom	Survival Horror	2021	84	8.50	100000
9	10	Resident Evil (Remake)	Capcom	Survival Horror	2002	91	1.35	0

Report generated by YData.

Overview: The dataset has 20 game entries with 8 complete variables and no missing data.

Variables: `id` and `name` are unique identifiers, while `developer` shows moderate variety.

Variables: `genre` is categorical, with Action-Adventure and Survival Horror being the most common.

Variables: `metacritic_score` ranges from 80 to 97, with an average of 90.2.

Variables: `sales_millions` varies significantly, including one game with zero sales.

Variables: `peak_players` has many zeros and one extreme high value of 6 million.

Interactions: The analysis will explore relationships between variables like sales and player count.

Correlations: High correlations are noted for `id` and `peak_players` with other variables.

Missing values: The dataset is complete with zero missing values across all columns.

Sample: The first ten rows show a focus on Call of Duty and Resident Evil franchises.

Conclusion

Successfully **create a data report analysis using data-profiling.**